BEHAVIOR THERAPY 29, 539–543, 1998

Behavioral Therapy Development and Psychological Science: If a Tree Falls in the Forest and No One Hears It . . .

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This special issue represents part of a continuing effort by the National Institute on Drug Abuse (NIDA) of the National Institutes of Health (NIH) to translate ideas and findings from basic behavioral science into behavioral therapy development research. The articles in this special issue are a product of the second in a series of workshops aimed at increasing the reciprocal exchange of ideas between behavioral therapy researchers and other behavioral scientists. This workshop series, "Behavioral Therapy Development and Psychological Science," began with a May 14–15, 1996, workshop. The papers from that workshop were published in a May 1997 special section of the journal *Psychological Science.* These articles are a product of a September 18–19, 1997, workshop. The purpose of both of these workshops is the same: reinforcing the bond between behavioral therapy development research and other, less applied, areas of psychology. Both workshops were spawned by NIDA's Behavioral Therapies Development Program,* a program of research aimed at comprehensively supporting all stages of behavioral therapy research, including the development of novel behavioral therapies.

As was the case for the May 1996 workshop, at the September 1997 workshop six behavioral scientists described their research, and six behavioral therapy researchers commented upon this research, and discussed how the

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*NIDA's program announcement, "Behavioral Therapies Development Program" (PA 94-078), announces support for all three stages of behavioral therapy development research, including: Stage I, or early therapy development, including research that translates basic behavioral research into early therapy development research; Stage II, including efficacy testing and research on the mechanism of action of therapies; and Stage III, or research on how to transport efficacious therapies to the community.

005-7894/98/0539–0543$1.00/0
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research presented may relate to behavioral therapy development. Thus, clinical researchers were paired with basic researchers. The pairing was done in an attempt to translate basic behavioral research paradigms into clinical research paradigms. It should be noted, of course, that once this is done, the research paradigm remains, indeed, a basic one, but it is now being utilized with a clinical population for clinical goals. That is, there are goals of not only understanding and explicating principles of behavior change, but also applying these principles therapeutically. Hence, such a "translated" basic behavioral research paradigm may become part of a series of experiments leading to the development of a new behavioral therapy, or the refinement of an existing one.

Also, the pairing was done in an attempt to mimic what might need to take place outside the workshop to increase communication across different communities of researchers so that therapy development research can make use of advances in basic research and so that basic research can be tested and influenced in the attempt to provide answers to questions dealing with clinical problems and populations. Although there exist researchers who conduct both basic behavioral and behavioral therapy research, this is a small group compared to the larger groups of researchers who define themselves on one side of the fence or the other. Collaborative relationships may need to be established for translation research to take place. Without such translation, the knowledge gained through basic behavioral science cannot be put to use. Although many of the workshop participants defined themselves as both basic and clinical researchers, they were asked to play one role or the other at the workshop. They were asked to either present basic research or to comment upon a basic research presentation in terms of its potential clinical application.

In this issue, the first paper is written by Warren Bickel, Gregory Madden, and Nancy Petry (1998). They present research on the behavioral economics of drug dependence, showing how drug-taking behavior is a function of both cost and available alternatives. G. Alan Marlatt and Jason Kilmer (1998) discuss the obvious relevance of Bickel et al.'s (1998) work to behavioral therapy development. Indeed, many behavioral economic principles have already been incorporated into existing behavioral therapies for drug addicts. Marlatt and Kilmer take clinical application a step further, offering a "consumer" approach to drug abuse treatment as a potentially fertile area for behavioral therapy development research.

Michael Dougher (1998) delineates the processes by which stimulus relations are formed; that is, the ways in which objects and symbols, such as words, come to be treated as equivalent, and the ways in which generalizations to other stimuli are formed. William Follette (1998), in discussing Dougher's research, points out that it may be more difficult to dissociate stimuli from equivalence relations in the real world than it is in the laboratory. He discusses Hayes's Acceptance and Commitment Therapy (Hayes et al., in press), a therapy based upon basic behavioral research on stimulus equivalence, and describes how it attempts to break up stimulus equivalence relations and how it differs conceptually from cognitive therapy.

Ian Gotlib and Elena Krasnoperova (1998) review the evidence of whether biases in cognitive processes such as attention and memory that focus on emotionally negative information serve as vulnerability factors for depression. They propose intervention strategies to reduce those biases as a means of preventing and treating depression. Richard Bootzin and Katherine McKnight (1998)* discuss how the clinical implications of this work might be explored. They also discuss conceptual issues related to the role of vulnerability factors and how to advance knowledge when attempting to develop interventions based on them.

Jill Hooley (1998) reviews the literature about the relationship of expressed emotion (EE; high levels of criticism, hostility, and emotional overinvolvement) to the long-term functioning of individuals with major mental disorders. Psychiatric patients who return from the hospital to families high in expressed emotion are significantly more likely to relapse than those who return to families low in expressed emotion. Hooley notes that the EE construct "has already been the catalyst for the development of a number of family-based interventions." Samuel Turner (1998) discusses the importance of carefully operationalizing a construct such as EE and evaluating its applications in each problem area thoroughly. In this way, knowledge can be accumulated about when and for whom interventions based on EE are likely to be most effective.

Peter Lang, Bruce Cuthbert, and Margaret Bradley (1998) have focused on emotion, the startle response, and pathological anxiety. They discuss research showing that, among anxiety-disordered individuals, the greater the level of psychopathology (e.g., PTSD and panic disorder with agoraphobia), the larger the basal startle response. Those individuals with less severe anxiety disorders (e.g., simple phobia or social phobia) show more cue-specific startle potentiation. The Lang et al.–Foa and Kozak (1998) set of papers is an ideal example of how basic behavioral science can be translated and utilized to develop novel and potent behavioral therapies. Edna Foa and Michael Kozak discuss how the research Lang and his colleagues present has been incorporated into their own research on the development of behavioral therapies for obsessive-compulsive disorder and PTSD. They also comment on how some of the more recent findings presented may impact upon the future development of behavioral therapies.

Elizabeth Loftus and Guiliana Mazzoni (1998) present their research on memory and imagination and the processes by which individuals may be led to remember events that, in fact, never occurred. They discuss research suggesting that the process of imagination may increase people's tendency to believe events actually happened, and highlight the concept of using imagination in a clinically useful way. Timothy Strauman (1998) examines the significance of this research within the broader context of research on social influence, motivation, and cognition. In his artful integration of diverse realms of psychology, he underscores the importance of bringing multiple scientific

*Neil Jacobson led the discussion of Ian Gotlib's presentation at the workshop but was unable to contribute a written commentary.

perspectives together, and he provides direction for a logical next step for clinical research on imagination and persuasion.

In his commentary on the papers in this special issue, Kenneth Howard (1998) provides a historical perspective that sheds light on the separation between applied and academic psychology. He also emphasizes that advances in clinical practice may depend more on our ability to match treatments to patients than in developing new treatments with the expectation that they will be effective for everyone. Thus, his essay makes a strong case for the need to develop and utilize research methodologies that can determine which treatments are best for particular individuals.

The introduction to the *Psychological Science* special section (Onken & Blaine, 1997), discussed how a major barrier had been lifted with regard to linking basic behavioral science to the development of behavioral therapies. That is, in 1992, NIDA released a program announcement ("Development of Theoretically Based Psychosocial Therapies for Drug Addiction") and began funding a cadre of researchers to carry out what is now called "Stage I" behavioral therapy development research, mostly under the "R01" regular research grant mechanism (investigators are now funded under NIDA's "Behavioral Therapies Development Program" announcement). The existence of this program means that researchers now may receive funding to translate ideas from basic behavioral research into research on the development and refinement of new and better behavioral therapies.

On most counts, NIDA's Stage I research program has been extraordinarily successful. To date, nearly 40 Stage I research projects have been funded. Researchers are developing therapies for individuals with cocaine, opiate, polydrug, or nicotine-use problems and also for individuals with substance-use disorders who have dual diagnoses of depression, schizophrenia, or personality disorder. Therapies are being developed specifically for females with addiction problems, for adolescents, for mothers, and for those who live in rural areas. NIDA places tremendous value on the potential inherent in the research currently being conducted to improve behavioral therapies for countless individuals, and has consistently demonstrated its interest in facilitating this research program.

As successful as this research program has been, however, there remains room for improvement. For while many therapies are being modified or refined for new populations, and while this is an extremely worthwhile pursuit, few investigators are pursuing the development of truly novel therapy approaches or modification of therapies, based on basic behavioral science discoveries or incorporating basic behavioral research paradigms. It should come as no surprise that the existence of potential funding to support a research endeavor does not guarantee that the research will be conducted. And, perhaps, this is where this special issue of *Behavior Therapy* and the prior special section of *Psychological Science* will play a role. Hopefully, some readers will be inspired to conduct Stage I behavioral therapy development research linking basic science with clinical science, based, at least in part, upon what they read here.

Psychology as a science can tell us countless things about our behavior, and how we might change it. This information, and the research paradigms from which the information was obtained, are there for the taking. Within this special issue are six examples of work that, potentially, could affect the way therapists do therapy. The trick is to take this work, and to use it within the context of a research project aimed at the development of a new therapy, or a therapy with a new twist. The trick is in the translation.

Of course, the process of science is an unpredictable one. And it may be that none of the things that we think will happen when we translate basic behavioral findings into clinical research and practice actually do happen. The beauty of it is that this, too, is informative. And this, too, can be put to good clinical use.

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RECEIVED: September 3, 1998
ACCEPTED: September 3, 1998